MUTATION IN BRIEF

Six Novel Alleles Identified in Italian Hereditary Fructose Intolerance Patients Enlarge the Mutation Spectrum of the Aldolase B Gene

Gabriella Esposito[1,2°], Rita Santamaria[1,2°†], Luigi Vitagliano[3], Luigi Ieno[1,2], Antonietta Viola[1,2], Laura Fiori[4], Giancarlo Parenti[5], Lucia Zancan[6], Adriana Zagari[2,3], and Francesco Salvatore[1,2]

[1] Dipartimento di Biochimica e Biotecnologie Mediche, Università di Napoli "Federico I", and [2] CEINGE-Biotecnologie Avanzate, Napoli, Italy; [3] Istituto di Biostrutture e Bioimmagini, CNR and Dipartimento di Chimica Biologica, Università di Napoli "Federico I", Napoli, Italy; [4] Istituti Clinici di Perfezionamento, Università di Milano, Milano, Italy; [5] Dipartimento Clinico di Pediatria Generale e Specialistica, Università di Napoli "Federico I", Napoli, Italy; [6] Dipartimento di Pediatria, Università di Padova, Padova, Italy

*Correspondence to: Prof. Francesco Salvatore, Dipartimento di Biochimica e Biotecnologie Mediche, CEINGE-Biotecnologie Avanzate, Università di Napoli "Federico II", Via S. Pansini 5, I-80131 Naples, Italy; Tel.: +(39) 081 7463135; Fax: +(39) 081 7463650; E-mail: salvator@unina.it

°These authors contributed equally to the work.
† Dipartimento di Farmacologia Sperimentale, Università di Napoli “Federico I”, Napoli, Italy

Grant sponsors: MIUR (DM 623/96; PS35-126/IND), Ministero della Salute (Progetti speciali, D.L.vo 299/99, bando 2001-02-03) and Regione Campania (Ricerca Sanitaria Finalizzata).

Communicated by Andreas Gal

Hereditary fructose intolerance (HFI) is a recessively inherited disorder of carbohydrate metabolism caused by impaired functioning of human liver aldolase (B isoform; *ALDOB*). To-date, 29 enzyme-impairing mutations have been identified in the aldolase B gene. Here we report six novel HFI single nucleotide changes identified by sequence analysis in the aldolase B gene. Three of these are missense mutations (g.6846T>C, g.10236G>T, g.10258T>C), one is a nonsense mutation (g.8187C>T) and two affect splicing sites (g.8180G>C and g.10196A>G). We have expressed in bacterial cells the recombinant proteins corresponding to the g.6846T>C (p.I74T), g.10236G>T (p.V222F), and g.10258T>C (p.L229P) natural mutants to study their effect on aldolase B function and structure. All the new variants were insoluble; molecular graphics data suggest this is due to impaired folding. © 2004 Wiley-Liss, Inc.

KEY WORDS: Aldolase B; *ALDOB*; hereditary fructose intolerance; HFI; protein expression; molecular modeling

INTRODUCTION

Fructose-1,6-bisphosphate aldolase (EC 4.2.1.13) is a homotetrameric enzyme that catalyses the reversible aldol cleavage of fructose 1,6-bisphosphate (FBP) and fructose 1-phosphate (F1P). The three mammalian aldolase isoenzymes are tissue-specific: muscle (aldolase A), liver (aldolase B) and brain (aldolase C) [Salvatore *et al.*, 1986]. In humans, aldolase B deficiency causes hereditary fructose intolerance (HFI, MIM# 229600), a potentially

Received 10 May 2004; accepted revised manuscript 25 August 2004.

© 2004 WILEY-LISS, INC.
DOI: 10.1002/humu.9290

lethal recessively inherited disorder of carbohydrate metabolism whose frequency is estimated to be 1 in 20,000 births [Steinmann *et al.,* 2001]. Cases of HFI are constantly increasing because of increased sugar consumption in industrialized countries. In fact, the enhanced incidence of HFI, which is a potentially fatal disease of infants and young children, has been attributed to changes in maternal behavior in wealthy nations [Cox, 2002]. Therefore, it would be more effective, in terms of prevention, to identify individuals who are maladapted to modern diets rather than attempting to counter the sugar industry.

The human aldolase B gene, located on chromosome 9q22.3, is 14,500-bp long (GenBank Accession #NT_008470.16), contains nine exons and encodes a 364-amino acid polypeptide (type B monomer, GenBank Accession #NP_000026.2). To date, 29 mutations have been identified in the aldolase B gene of HFI patients, as reported by Steinmann et al. [2001] and in the Human Gene Mutation Database [Stenson et al., 2003]; these include missense and nonsense mutations, large and small deletions and mutations in the splicing regions. The most common HFI-mutations in Europe are c.448G>C (p.A150P) and c.524C>A (p.A175D) in exon 5 (trivial names A149P and A174D, respectively), which account for about 85% of all HFI alleles. In particular, the most frequent p.A150P allele has a wide distribution among populations in Europe, including Ashkenazi Jews and other residents of Israel, Turkey, North America, The Dominican Republic, Australia and New Zealand [Ali et al., 1998]. Haplotype analysis with informative polymorphisms in aldolase B gene provides evidences that both p.A150P and p.A175D mutations derived from single ancestral events and have spread by mass migration and genetic drift [Brooks and Tolan, 1993]. The other mutations are spread throughout the aldolase B gene, and their frequency differs among ethnic groups [Steinmann et al., 2001]: some of these are now widespread as result of increasing inter-marriage and genetic admixture, other are private mutations [Ali et al., 1998].

We previously reported known and unknown disease-causing mutations in the aldolase B gene of patients with clinical symptoms of HFI [Santamaria *et al.,* 1993; Santamaria *et al.,* 1996; Santamaria *et al.,* 1999; Santamaria *et al.,* 2000]. Here we describe six new aldolase B gene mutations identified in seven unrelated HFI Italian patients. Molecular graphics data indicate that these mutants perturb enzyme function and structure.

MATERIALS AND METHODS

Patients

We looked for aldolase B gene mutations in the aldolase B gene of seven unrelated Italian patients affected by clinical symptoms suggestive of HFI. Data of fructose challenge and aldolase activity on liver biopsy were available for some patients.

Mutation analysis

Genomic DNA was extracted from blood samples with the Nucleon BACC2 kit (Amersham Pharmacia Biotech UK Ltd.). Mutation numbering is based on gDNA (GenBank Accession #NT_008470.16), cDNA (#NM_000035.2) and protein (#NP_000026.2) sequences; for cDNA numbering, +1 corresponds to the A of the ATG translation initiation codon, which is the codon 1. We used the amplification refractory mutation system (ARMS) [Newton *et al.,* 1989] to test patients' DNA for the presence of seven known HFI mutations – c.448G>C (p.A150P), c.524C>A (p.A175D), c.356_359delCAAA (p.N120K121del), c.672T>A (p.Y204X), c770T>C (p.L257P), c.1005C>G (p.N335K), c.1013C>T (p.A338V) - which account for about 95% of HFI Italian alleles. Allele-specific ARMS primers, and common and control primers are reported in Table 1. The oligonucleotides used to amplify the nine exons of the aldolase B gene, their splice junctions and the promoter region are reported elsewhere [Santamaria *et al.,* 1993]. We sequenced the PCR products with the Big Dye™ Terminator Cycle Sequencing kit and the ABI Prism 377 Genetic analyzer (PE Applied Biosystems, Foster City, CA, USA). Novel mutations detected by sequencing were confirmed with a different method (i.e. restriction analysis, ARMS, see below) and on a second sample. To verify that the novel missense mutations are not neutral polymorphisms, we checked samples from at least 100 normal individuals for each of the novel missense mutations. We screened the g.6846T>C (p.I74T) allele using PCR-RFLP and *Msp I* digestion because the T>C transition creates an additional cleavage site. We used ARMS to screen the g.10236G>T (p.V222F) and g.10258T>C (p.L229P) alleles, and PCR-RFLP with *Dde I* digestion to follow the Mendelian inheritance of the g.8187C>T (p.Q111X) and g.8180G>C (c.325-1G>C) in the probands' families because both mutations create a new site for this restriction endonuclease.

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Primers Used for ARMS Analysis

ARMS primer	Sequence	Common primer
150 Norm	5'-GTGTTGACTTTGGGAAGTGGGGT**G**-3'	5'-GAAGAAAACTCTAGCCTACT-3'
150 Mut	5'-GTGTTGACTTTGGGAAGTGGGGT**C**-3'	
175 Norm	5'-AGCACCTGCTGACAGATGGTG**G**-3'	5'-TAGAAGCCCCATGGATCAGG-3'
175 Mut	5'-GAGCACCTGCTGACAGATGGTG**T**-3'	
MD4 Norm a	5'-GAGGTGCTCCTCTTGCAGGAAGAAA**C**-3'	5'-GCCTTCATTTCTAGCTTAC-3'
MD4 Mut a	5'-GAGGTGCTCCTCTTGCAGGAAGAAA**G**-3'	
204 Norm	5'-CCATGACCTGGAACACTGCCACTA**T**-3'	5'-AGCTGTTATATGTTAAGTAACA -3'
204 Mut	5'-CCATGACCTGGAACACTGCCACTA**A**-3'	
257 Norm	5'-GTAGCTATGGCCACCGTAACAGGTC**T**-3'	5'-AAGAAAACAATGCTTCTCCG-3'
257 Mut	5'-TAGCTATGGCCACCGTAACAGGTC**C**-3'	
335 Norm	5'-CTCCTCTCTCATAAAGTCTTCTAGGGTAA**C**-3'	5'-TGTCGAATTTCCAGGATTAG-3'
335 Mut	5'-CTCCTCTCTCATAAAGTCTTCTAGGGTAA**G** -3'	
338 Norm	5'-TCTTCTAGGGTAAGCTAACTGCGAGG**C**-3'	
338 Mut	5'-GTCTTCTAGGGTAAGCTAACTGCGAGG**T**-3'	
Control $_A$b	5'-CCTTGGACCCAGAGGTTCTT-3'	
Control $_B$b	5'-AATCATTCGTCTGTTTCCCT-3'	

-globin [a] to detect the c.356_359delCAAA mutation; [b]within the gene

Construction and expression of mutant aldolase B

We used the full-length wild type human aldolase B cDNA, previously cloned into the His-tag bacterial expression vector pET-16b [Santamaria *et al.*, 2000] as a template to generate the aldolase B missense mutants p.I74T, p.V222F and p.L229P according to the instructions in the QuickChange™ site-directed mutagenesis kit (Stratagene Inc.). Wild-type and mutated recombinant plasmids were purified over Qiagen columns (Qiagen Inc., Valencia, CA, USA), completely sequenced and used to transform the *E. coli* BL21(DE3) cells as described previously [Esposito *et al.*, 2002].

Molecular graphics analyses

The mutation-induced structural alterations were studied by using the crystal structure of aldolase B (PDB code 1QO5) as a reference [Dalby *et al.*, 2001]. The analyses were conducted with the PROMOTIF [Hutchinson *et al.*, 1996], O [Jones *et al.*, 1997)] and PROCHECK [Laskowski *et al.*, 1993] programs. Figure 2A and B was generated with the MolScript [Kraulis *et al.*, 1991] and Raster3D [Merritt *et al.*, 1997] programs.

RESULTS

We have set-up an ARMS analysis that rapidly detects seven common disease-causing mutations (p.A150P, p.A175D, p.N120K121del, p.Y204X, p.L257P, p.N335K and p.A338V), accounting for about 95% of HFI alleles in the Italian population. In the seven HFI patients reported here, ARMS analysis established that all but one were heterozygous for one of these frequent mutations. To identify the second HFI allele, we sequenced the whole aldolase B gene coding region, the splicing junctions and the promoter region, and detected six new nucleotide changes. Three of these were missense mutations, one was a nonsense mutation and two affect splicing sites. The mutations identified are reported in Table 2.

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Six Novel Aldolase B Mutations Identified in HFI Patients

Exon	Nucleotide change	Effect
3	g.6846T>C	p.I74T
4	g.8180G>C (c.325-1G>C)	deduced splicing defect
4	g.8187C>T	p.Q111X
7	g.10196A>G (c.625-2A>G)*	deduced splicing defect
7	g.10236G>T	p. V222F
7	g.10258T>C	p.L229P

Numbering according to GenBank Accession # NT_008470.16 (gene); # NP_000026.2 (protein) and # NM_000035.2 (cDNA); for cDNA numbering, +1 corresponds to the A of the ATG translation initiation codon which is the codon 1; *detected in two unrelated HFI patients.

An adult female with clinical symptoms of HFI in which ARMS analysis did not reveal any of the common HFI mutations, was homozygous for the novel missense p.I74T mutation (Fig. 1 a). Because it was impossible to check the Mendelian inheritance of this mutation in the patient's family, we performed southern blot analysis and did not find fragments suggestive of gross rearrangements within the aldolase B gene (data not shown). In a patient heterozygous for the common p.N335K allele, we found a C>T transition in exon 4 that changes the Gln residue at position 111 into a stop codon (Fig. 1b). This mutation led to a truncated protein lacking residues essential for substrate binding. Mendelian inheritance was demonstrated in the proband's family with restriction analysis using *Dde I* digestion.

The g.8180G>C (c.325-1G>C) mutation (Fig. 1c) alters the canonical splice acceptor site of IVS3 and was found in a six-year-old patient in association with the widespread p.A150P mutation. Also in this family, Mendelian inheritance was demonstrated by *Dde I* digestion. The g.10196A>G (c.625-2A>G) splicing mutation (Fig. 1 d) was detected in two unrelated patients, compound heterozygous for the p.Y204X and p.A150P mutations, respectively. Both mutations alter canonical splice acceptor sequences and thereby they should prevent proper maturation of the primary transcript, as already reported for different mutations in the acceptor splicing site of both IVS3 (g.1133G>A) and IVS6 (g.10197G>A) [Sanchez-Gutierrez *et al.*, 2002; Ali *et al.*, 1994].

The last two novel mutations, p.V222F and p.L229P (Fig. 1e and f, respectively), both located in exon 7, lead to amino acid changes within the same protein region. They were found in two patients, both heterozygous for the widespread p.A175D mutation. ARMS analysis demonstrated Mendelian inheritance in the two families.

None of the new HFI natural variants reported herein, p.I74T, p.V222F and p.L229P, was found in 100 normal unrelated subjects analyzed as controls. To verify the effects of these missense mutations on enzyme structure and function, we performed site-directed mutagenesis and expressed the recombinant mutated proteins in *E. coli* cells using a procedure previously used for aldolase B [Esposito *et al.*, 2002] and aldolase A mutants [Esposito *et al.*, 2004]. Like the HFI-related mutants p.A175D and p.L183_V184del [Rellos *et al.*, 2000; Esposito *et al.*, 2002], recombinant proteins corresponding to p.I74T, p.V222F and p.L229P were insoluble, since they precipitated in the *E. coli* inclusion bodies at different temperatures of bacterial growth and induction (37°C or 22°C).

DISCUSSION

Hereditary fructose intolerance is an inborn error of carbohydrate metabolism caused by a deficiency of aldolase B. The clinical manifestations of HFI, which occur after ingestion of fructose, sucrose or sorbitol (the last two are potential sources of dietary fructose), consist in abdominal pain, vomiting and such metabolic disturbances as acidosis and hypoglycemia that may be fatal. Symptoms and signs usually regress once fructose is totally excluded from the diet [Steinmann *et al.*, 2001].

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

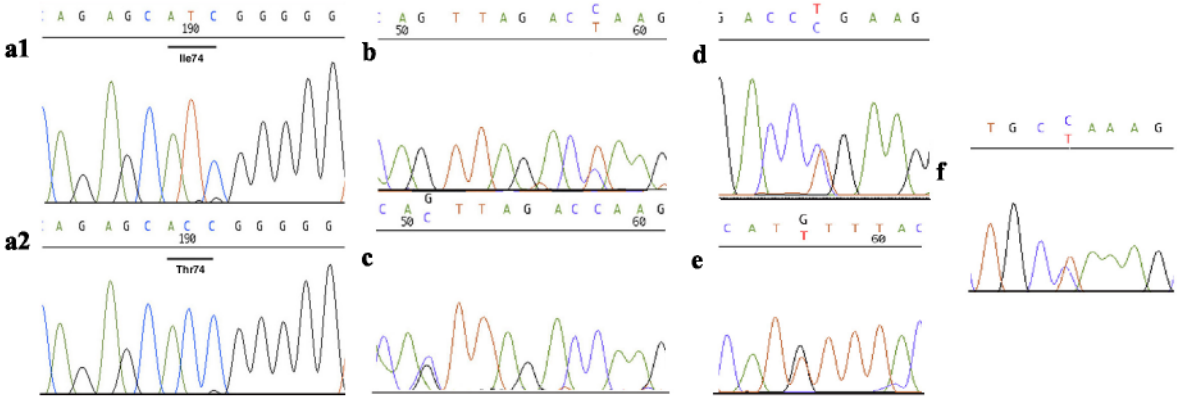


Figure 1. Sequence electropherograms showing the aldolase B gene regions with the novel mutations found in HFI patients. **a1:** normal exon 3 sequence at the p.I74 position; **a2:** p.I74T mutation (T>C) at the homozygous state; **b:** p.Q111X (C>T), exon 4; **c:** c.325-1 G>C, beginning of exon 4.; **d:** c.625-2A>G (T>C on the antisense strand), beginning of exon 7; **e:** p. V222F (G>T), exon 7; **f:** p.L229P (T>C), exon 7. Mutations from **b.** to **f.** were at the heterozygous state.

DISCUSSION

Hereditary fructose intolerance is an inborn error of carbohydrate metabolism caused by a deficiency of aldolase B. The clinical manifestations of HFI, which occur after ingestion of fructose, sucrose or sorbitol (the last two are potential sources of dietary fructose), consist in abdominal pain, vomiting and such metabolic disturbances as acidosis and hypoglycemia that may be fatal. Symptoms and signs usually regress once fructose is totally excluded from the diet [Steinmann *et al.*, 2001].

The procedures used to diagnose HFI, i.e., measurement of blood variables after intravenous fructose challenge and of aldolase B activity in fresh biopsy samples of liver tissue, are invasive and do not reveal the heterozygous state. Differently, molecular analysis to identify disease-causing mutations in the aldolase B gene is non-invasive and accurate also in diagnosing heterozygous subjects.

In this study, we have set-up the conditions for the rapid ARMS screening of seven HFI-mutations, which correspond to about 95% of the HFI-alleles in Italy. The remaining 5% account for rare or unknown mutations, often found in association with a common HFI-allele. In contrast to the USA and European populations, where the p.A150P and p.A175D mutations account for at least 85-90% of the HFI-alleles, Italian patients proved to be more heterogeneous. In fact, the two widespread mutations cover approx. 65% of the disease-alleles, while the remaining 30% is represented by mutations c.356_359delCAAA, p.Y204X, p.L257P, p.N335K and p.A338V, which are less frequent in other populations. The ARMS analysis reported here allows for the rapid, accurate identification of the HFI causative mutations in our population at a low cost.

In seven unrelated patients, all but one heterozygous for a mutation analyzed by ARMS, direct sequencing of the whole aldolase B gene revealed six novel point mutations. These novel mutations, resulting in aldolase B deficiency, seem to be rare or private.

Mutations c.325-1 G>C and c.625-2A>G (the latter present in two unrelated patients) affect the acceptor splicing site in intron 3 and 6, respectively. To date, five splicing defects have been reported in the aldolase B gene of HFI patients, two of which specifically affect the invariable canonical dinucleotide sequence in the intron-exon boundaries [Cross *et al.*, 1990; Brooks *et al.*, 1991; Ali *et al.*, 1994; Ali *et al.*, 1996, Sanchez-Gutierrez *et al.*, 2002]. We did not test experimentally the consequences of c.325-1 G>C and c.625-2A>G on RNA processing: however, as

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

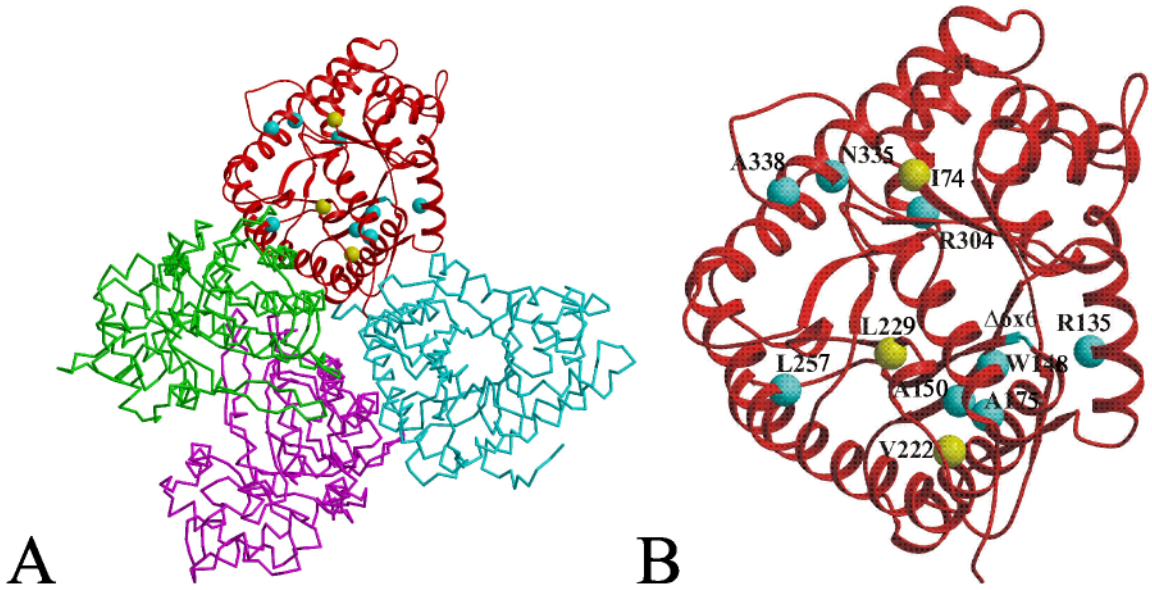


Figure 2. Location of HFI-related missense mutations in the structure of the native aldolase B. Panel A: the aldolase B tetramer structure. Panel B: a close-up view of a single chain of the tetramer. The three novel missense mutations reported herein, are 6ex6) coloured in yellow. The site of the p.L183_V 184del mutation (designated Santamaria *et al.*, 1999) is also shown.

observed for other disease-related genes [Krawczac *et al.*, 1992; Nakai *et al.*, 1994], the mutations may dramatically affect the correct reading frames thereby giving rise to null-alleles. Similarly, the novel p.Q111X mutation may result in a null-allele. p.Q111X mutation falls at the beginning of exon 4 and produces a truncated protein that lacks the substrate binding residues Lys147, Arg149 and Lys 230 (which form the Schiff base), lined within the active site pocket. Four other nonsense mutations that cause null-alleles have been identified in the aldolase B gene of HFI patients [Ali *et al.*, 1993; Ali *et al.*, 1994; Kajihara *et al.*, 1990]. One of these, p.Y204X, that produces a protein 93 amino acids longer than that derived from the Q111X allele, represents 10% of the HFI alleles in the Italian population.

The other single-nucleotide mutations reported here give rise to the natural aldolase B variants p.I74T, p.V222F and p.L229P. Residues Ile 74, Val 222 and Leu 229 are conserved in most aldolase B sequences and they are either conserved or replaced with similar amino acids in the sequences of aldolase A and C isozymes. To evaluate the effect of these mutations on enzyme function, we reproduced them in bacterial cells by site-directed mutagenesis. All the recombinant mutated proteins precipitated in the *E. coli* inclusion bodies, which is indicative of severe folding alterations. This conclusion is corroborated by molecular modeling analysis demonstrating that the three missense mutations significantly alter the enzyme structure. In the case of p.L229P, the insertion of a proline - strand residue in position 229 destabilizes the 228-230 and the hydrophobic pocket constituted by the polar side chains of Leu 229, Val 261 and Pro 189.

Mutation p.I74T is also expected to cause considerable structural perturbations. Ile 74 is located in the 73-80 and its side-chain interacts with the Ala 332, Leu 306 and Val 77 side chains. The Ile>Thr replacement destabilizes enzyme structure because of the different size and polarity of the two residues. Finally, Val 222 is located in a hydrophobic pocket formed by Val 217, Ala 175 and the C^2 atom of Thr 227. The replacement with the bulky side-chain of a Phe residue destabilizes the region, which is also close to the tetramer interface. Notably, p.V222F is the first HFI-related missense mutation not to be located in a secondary structure element of aldolase B.

In conclusion, our findings notably enlarge the spectrum of HFI-related mutations reported thus far and indicate other residues as important for correct enzyme folding.

ACKNOWLEDGMENTS

We are indebted to Jean Ann Gilder for editing the text.

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

REFERENCES

Ali M, Rosien U, Cox TM. 1993. DNA diagnosis of fatal fructose intolerance from archival tissue. Quart J Med 86:25-30

Ali M, Tuncman G, Cross N, Vidailhet M, Bokesoy I, Gitzelmann R, Cox TM. 1994. Null alleles of the aldolase B gene in patients with hereditary fructose intolerance. J Med Genet 31:499-503

Ali M, James CL, Cox TM. 1996. A newly identified aldolase B splicing mutation (G>C, 5' intron 5) in hereditary fructose intolerance from New Zeland. Hum Mutat 7:155-157

Ali M, Rellos P, Cox T. 1998. Hereditary fructose intolerance. J Med Genet 35:353–365

Brooks CC, Buist N, Tuerck,J, Tolan DR. 1991. Identification of a splice site mutation in the aldolase B gene from an individual with hereditary fructose intolerance. Am J Hum Genet 49:1075-81

Brooks CC, Tolan DR. 1993. Association of the widespread A149P hereditary fructose intolerance mutation with newly identified sequence polymorphism in the aldolase B gene. Am J Hum Genet 52:835-840

Cox TM. 2002. The genetic consequences of our sweet tooth. Nat Rev Genet 3:481-487

Cross NC, Cox TM. 1990. Partial aldolase B gene deletion in hereditary fructose intolerance. Am J Hum Genet 47:101-106

Dalby RD, Tolan DR, Littlechild JA. 2001. The structure of human liver fructose-1,6-bisphosphate aldolase. Acta Cryst D Biol Crystallogr. 57:1526-1533

Esposito G, Vitagliano L, Santamaria R, Viola A, Zagari A, Salvatore F. 2002. Structural and functional analysis of aldolase B mutants related to hereditary fructose intolerance. FEBS Lett, 531:152-156

Esposito G, Vitagliano L, Costanzo P, Borrelli L, Barone R, Pavone L, Izzo P, Zagari A, Salvatore F. 2004. Human aldolase A natural mutants: relationship between flexibility of the C-terminal region and enzyme function. Biochem J 380:51-56

Hutchinson EG, Thornton JM. 1996. PROMOTIF -- a program to identify and analyze structural motifs in proteins. Protein Sci 5:212–220

Jones TA, Kjeldgaard M. 1997. Electron-density map interpretation. Methods Enzymol. 277:173–208

Kajihara S, Mukai T, Arai Y, Owada M, Kitagaua T, Hori K. 1990. Hereditary fructose intolerance caused by a non sense mutation of the aldolase B gene. Am J Hum Genet 47:562-567

Kraulis PJ. 1991. MolScript: a program to produce both detailed and schematic plots of protein structures. J Appl Crystallogr 24:945–949

Krawczac J, Reiss J, Cooper DN. 1992. The mutational spectrum of single base-pair substitutions in mRNA splice junctions of human genes/ causes and consequences. Hum Genet 90:41-54

Laskowski RA, MacArthur MW, Moss DS, Thornton JM. 1993. PROCHECK: a program to check the stereochemical quality of protein structures. J Appl Cryst 26:283–291

Merritt EA, Bacon DJ. 1997. Raster3D: photorealistic molecular graphics. Methods Enzymol 277:505–524

Nakai K, Sakamoto H. 1994. Construction of a novel database containing aberrant splicing mutations of mammalian genes. Gene 141:171-177

Newton CR, Graham A, Heptinstall LE, Powell SJ, Summers C, Kalsheker N, Smith JC, Markham AF. 1989. Analysis of any point mutation in DNA. The amplification refractory mutation system (ARMS). Nucleic Acids Res 17:2503-2516

Rellos P, Sygusch J, Cox TM. 2000. Expression, purification and characterization of natural mutants of human aldolase B. J Biol Chem, 275:1145-1151

Salvatore F, Izzo P, Paolella G. 1986. Aldolase gene and protein families: structure, expression and pathophysiology. In: Blasi F, editor. Horizons in Biochemistry Biophysics, John Wiley and Sons. 8:611–665

Sanchez-Gutierrez JC, Benlloch T, Leal MA, Samper B, Garcia-Ripoll I, Feliu JE. 2002. Molecular analysis of the aldolase B gene

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in patients with hereditary fructose intolerance from Spain. J Med Genet 39:e56

Santamaria R, Scarano M I, Esposito G, Chiandetti L, Izzo P, Salvatore F. 1993. The molecular basis of hereditary fructose intolerance in Italian children. Eur J Clin Chem Clin Biochem 31:675–678

Santamaria R, Tamasi S, Del Piano G, Sebastio G, Andria G, Borrone C, Faldella G, Izzo P, Salvatore F. 1996. Molecular basis of hereditary fructose intolerance in Italy: identification of two novel mutations in the aldolase B gene. J Med Genet 33:786–788

Santamaria R, Vitagliano L, Tamasi S, Izzo P, Zancan L, Zagari A, Salvatore F. 1999. Novel six-nucleotide deletion in the hepatic fructose-1,6-bisphosphate aldolase gene in a patient with hereditary fructose intolerance and enzyme structurefunction implications. Eur J Hum Genet 7:409–414

Santamaria R, Esposito G, Vitagliano L, Race V, Paglionico I, Zancan L, Zagari A, Salvatore F. 2000. Functional and molecular modelling studies of two hereditary fructose intolerance-causing mutations at arginine 303 in human liver aldolase. Biochem J 350:823–828

Steinmann B, Gitzelmann R, Van den Berghe G. 2001. Disorders of fructose metabolism. In: Scriver CR, Beaudet AL, Valle D, Sly WS, editors. The metabolic and molecular bases of inherited disease, 8th ed. New York:McGraw-Hill. p 1489-1520

Stenson PD, Ball EV, Mort M, Phillips AD, Shiel JA, Thomas NS, Abeysinghe S, Krawczak M, Cooper DN. 2003. The Human Gene Mutation Database (HGMD): 2003 Update. Hum Mutat 21:577-581.

10981004, 2004, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/humu.9290 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License